

PC-AVCT: Context-Augmented, Session-Calibrated Blood Pressure Tracking from PPG

Farouk Ganiyu Adewumi and Timothy Oladunni

Abstract—Cuffless blood pressure (BP) estimation from photoplethysmography (PPG) is vulnerable to participant leakage, ambiguous calibration burden, and protocol shortcuts. We evaluated PC-AVCT, a context-augmented implementation motivated by Attractor-Vascular Coupling Theory (AVCT), for session-anchored systolic (SBP) and diastolic (DBP) tracking. The primary retrospective analysis used three repeats of five-fold participant-level internal cross-validation on 653 AuroraBP participants, excluding calibration-labelled rows from fitting and scoring. With a three-reading cuff anchor, the rerun wearable-state model achieved participant-level MAE of 7.697/6.139 mmHg. Paired gains over the same core without added context were 0.436 mmHg SBP (97.5% CI 0.305–0.573) and 0.126 mmHg DBP (0.053–0.200): statistically resolved but small relative to clinical accuracy requirements. In a separate 108-participant pulse-rate study, quality-weighted visible/infrared fusion achieved 3.690 BPM MAE. A post-hoc comparison against equal weighting (3.985 BPM) showed a 0.294 BPM improvement (0.069–0.567); guarded skin-tone correction did not improve overall error. A separate VitalDB implementation, fitted using 55 development subjects, was evaluated on 228 cases from 227 disjoint held-out subjects and 5,472 future windows. A fixed 50/50 blend of context and context-plus-PPG predictions improved over calibration, time, and demographics by 0.451 mmHg SBP (0.121–0.781) and 0.357 mmHg DBP (0.194–0.512), reaching 15.375/8.833 mmHg. These results support incremental context and PPG contributions in the evaluated settings, with separate evidence for pulse-rate fusion. They do not isolate quality-dependent BP weighting, demonstrate transfer of the fitted AuroraBP model, or establish cuff replacement, equitable performance, prospective validation, or standards compliance.

Index Terms—Photoplethysmography, cuffless blood pressure, nonlinear time-series features, one-session calibration, quality-aware fusion, pigmentation-intervention audit, data leakage, participant-level cross-validation, wearable sensing.

I. Introduction

A. Motivation

PPG-based cuffless BP estimation is attractive because optical sensors are already present in consumer wearables. Its evaluation must distinguish participant overlap, calibration burden, and acquisition-context shortcuts. Record-level splits can inflate apparent accuracy when correlated measurements from one participant enter both training and testing [1]. A model anchored to three initial cuff readings also answers a different question from one calibrated once or repeatedly supplied with recent cuff values. Finally, pigmentation-associated optical differences motivate intervention testing, but a small subgroup

or a nonsignificant contrast cannot establish equitable performance [2]–[4]. We audit anchoring, context, and pigmentation-related interventions using recorded Fitzpatrick labels. Candidate interventions and context arms were frozen before repeated participant-level validation, after earlier development on AuroraBP. Because a fixed holdout was inspected across successive ablations, it is retained as historical development evidence. The primary repeated analysis confines feature ranking and fitting to outer training participants but remains internal validation. Secondary negative results are reported without interpreting them as proof of equity, equivalence, or general failure of pigmentation-aware modeling.

B. What “PC-AVCT” Names

We retain PC-AVCT for continuity with the author-developed AVCT formulation [5]. Here it denotes a context-augmented, session-calibrated tracking framework with a secondary pigmentation-intervention audit. Three implementations must be distinguished: AuroraBP evaluates added context, the multi-wavelength study evaluates pulse-rate fusion, and VitalDB evaluates a separately fitted context-plus-PPG model and a fixed blend. Quality-dependent wavelength weighting is implemented in the pulse-rate study; it is not the VitalDB blending rule. The optional MST-dependent pulse-fusion route uses an exploratory inner-validation guard. These implementations do not constitute a single unchanged model transported between datasets.

C. Hypotheses

The development program considered five hypotheses, summarized here with a narrowed anchor-specific H5. This revised wording is retrospective and must not be interpreted as a prospectively specified hypothesis set. Historical model-development ablations used a fixed 501/168 split; the frozen final comparison was subsequently assessed by repeated participant-level outer cross-validation:

- H1 A one-session calibration anchor, combined with AVCT attractor features, tracks within-session SBP/DBP change more accurately than calibration retention (predicting no change).
- H2 Acquisition-context covariates improve tracking under a session anchor, with a separately tested conservative arm restricted to wearable-available posture, activity, and optical quality.

F. G. Adewumi and T. Oladunni are with the Department of Computer Science, Morgan State University, Baltimore, MD 21251, USA. E-mails: fagan1@morgan.edu; timothy.oladunni@morgan.edu.

- H3 Pigmentation-corrected optical features and explicit Fitzpatrick-aware model terms improve accuracy and/or narrow the between-skin-tone error gap relative to a reference model.
 - H4 Rebalancing the training population across Fitzpatrick bands improves fairness (narrows the between-band MAE gap) without harming overall accuracy.
 - H5 A same-session “most-recent-prior” cuff anchor can account for a large part of apparent improvement relative to session-start anchoring.
- H1 and H2 are supported for the evaluated models, and the historical anchor comparison supports the narrower H5. These comparisons do not isolate the attractor feature bank from all other inputs. The specific interventions tested under H3 and H4 are not supported: overall pigmentation-correction benefit is not established and the combined rebalancing/group-gating intervention increases error in the historical matched ablation. The small Fitzpatrick V–VI subgroup prevents interpreting those null or adverse intervention results as a general falsification of pigmentation-aware modeling or as evidence of equity.

D. Contributions

- 1) Participant-separated internal evaluation of added wearable-state context under a three-reading session anchor, with a protocol-enriched comparator and matched calibration-dose analysis.
- 2) A calibrated-tracking audit that separates initial calibration from recent-cuff anchoring and reports secondary pigmentation-intervention results without fairness claims.
- 3) Independent pulse-rate fusion evaluation and a separately fitted VitalDB implementation that tests incremental PPG utility beyond calibration, time, and demographics. The equal-weight pulse comparison was added post hoc during manuscript audit.

The main text presents the primary comparisons and independent-dataset results. Historical development ladders, auxiliary diagnostics, and exploratory endpoint specialization are retained in the supplementary appendices.

II. Related Work

A. PPG-Based Cuffless Blood Pressure Estimation

Cuffless BP estimation from PPG spans pulse-transit-time methods, hand-engineered waveform features, and end-to-end deep networks [6], [7]. Calibration-free approaches attempt to avoid any subject-specific reference [8], typically at a cost in accuracy relative to calibrated methods; single-point-calibrated approaches instead accept one reference reading and model change from it [5], [9]. Recommendations for evaluating PPG-based BP algorithms increasingly emphasize AAMI/ISO 81060-2 and BHS grading protocols and honest reporting of what these standards do and do not certify for calibrated, per-participant models [10].

B. Nonlinear and Information-Theoretic Feature Representations

The feature bank rests first on established time-series methods. Delay-coordinate reconstruction provides a generic mathematical basis for representing state from a scalar observation under specific dynamical assumptions [11]; it does not by itself prove that a finite PPG window recovers a unique cardiovascular state. Recurrence plots and recurrence-quantification analysis provide established descriptors of reconstructed trajectories [12], [13], while sample entropy and permutation entropy were developed as general complexity measures, including for short physiological series [14], [15]. PPG itself reflects cardiac-synchronous peripheral blood-volume change modulated by vascular, autonomic, respiratory, thermoregulatory, contact, and optical factors [16]. Independent work has specifically applied attractor reconstruction to arterial pulse-wave morphology [17].

CST, AVCT, and ADT are recent author-developed, hypothesis-generating formulations layered on these established methods [5], [18], [19]. They motivate the particular feature grouping, CSI term, calibration-delta representation, and terminology used here, but they are not treated as independently validated physiological laws, and their axioms are not required for the statistical validity of the empirical comparisons. Accordingly, this study tests whether the resulting feature architecture generalizes under participant-level validation; it does not claim that successful prediction proves the theories. A residualized mutual-information ranking restricts the candidate feature set rather than searching combinatorially over all features [20], [21].

C. Skin Pigmentation and PPG-Derived Vitals

Pigmentation-associated optical differences motivate evaluating both signal processing and downstream error [2]–[4], [22], [23]. A Nigerian rPPG field study enrolled 306 Fitzpatrick V/VI participants, obtained 249 analyzable estimates, and reported 15.4/10.9 mmHg SBP/DBP MAE with poor hypertension sensitivity [24]. Its lack of an I–IV comparator prevents attributing error specifically to skin tone. A small polarization-sensitive PPG study improved perfusion index in eight analyzed participants but did not estimate BP [23]. Such optical findings and gain-calibration methods [25] motivate interventions; they do not establish BP-tracking benefit. We therefore distinguish signal-quality associations, pulse-rate fusion, and BP error, and avoid interpreting nonsignificant subgroup differences as parity.

D. Evaluation Rigor and Data Leakage

The original AuroraBP study already emphasized strong calibration-based benchmarks and differences between laboratory and ambulatory performance [26]. The present contribution is the specific context, calibration-dose, and intervention comparisons on the analyzed subset; it is not the introduction of calibration-aware benchmarking itself.

A benchmark study across four public PPG-BP datasets (Sensors, UCI, BCG, PPGBP) explicitly compares record-level-leaked and subject-independent splits and reports that leakage alone can inflate apparent performance by more than 40 percentage points on some metrics [1], corroborating a broader concern in the deep-learning BP literature that record-level or window-level splits can substantially inflate reported accuracy relative to subject-independent evaluation. The MIMIC-BP data descriptor addresses this reproducibility problem by providing 1,524 subject-identified cases, balanced per-participant segments, and participant-disjoint training, validation, and test lists [27]; it is cited as a relevant external resource but was not used in the primary or endpoint-specialization results reported here. A separate line of work interrogates the information content of the PPG waveform itself, asking how much of reported model accuracy reflects genuine physiological signal versus dataset-specific shortcuts [28], [29]. Our calibration-row leakage audit (Section III-E) is a distinct but related artifact: not a train/test split violation, but a same-session near-zero-delta artifact specific to change-prediction (rather than absolute-value) BP models, and we are not aware of a prior study that isolates it.

III. The PC-AVCT Framework

A. Attractor Feature Bank

Let x_n , $n = 1, \dots, T$, denote a discretely sampled PPG window at rate f_s . Following delay-coordinate embedding [11], we reconstruct the trajectory

$$\mathbf{X} = \tau \mathcal{G}[x], \quad \mathbf{X}_{n,\cdot} = [x_n, x_{n+\tau}, \dots, x_{n+(d-1)\tau}], \quad (1)$$

with embedding dimension $d = 4$ and delay $\tau = 5$ samples, matching the AVCT/CST instantiation [5], [18]. From $\mathbf{X}$ we compute the shared PPG attractor feature bank

$$\Phi(x) = [\sigma_m, \text{skew}_m, \text{kurt}_m, H_{\text{SE}}, H_{\text{PE}}, R_{\text{rec}}, R_{\text{det}}, H_{\text{RQA}}, \text{CSI}, H_{\text{spec}}, Q] \quad (2)$$

comprising waveform moments (σ_m , skewness, kurtosis), sample and permutation entropy (H_{SE} , H_{PE}), recurrence rate and determinism (R_{rec} , R_{det}), recurrence entropy (H_{RQA}), CSI, spectral entropy, and optical quality. The auxiliary stress-test bank labels the unmodified measured waveform ref_^* , the simulated stressed waveform raw_^* , and its corrected version pc_^* ; “reference” does not mean latent physiological ground truth. AuroraBP instead extracts raw_^* from observed optical waveforms and pc_^* after the correction below. It does not construct an AuroraBP ref_^* input. The primary candidate pool contains raw-feature changes, time variables, and a UCI-derived prior; its exact provenance is described in Section IV-B.

B. Pigmentation Correction Operator

The implemented AuroraBP operator is a heuristic motivated by attenuation, rather than an estimate of melanin or a fitted Beer–Lambert law. Let $z = (\text{FST} - 1)/5$, $b = \text{median}(x)$, $a = \max(0.45, 1 - 0.55z)$, and $u = (x - b)/a$. With $W(u)$ a Wiener-filtered version

of u and $\lambda = \text{clip}(0.75z, 0, 0.75)$, the corrected channel is $b + (1 - \lambda)u + \lambda W(u)$. The filter kernel is an odd number of samples derived from $0.05f_s$, bounded between 5 and 21. A residual-to-bandpassed-signal SD ratio, clipped to $[0, 1.5]$, modifies entropy/recurrence tolerances; the corrected channel uses 0.35 times that ratio. The waveform median is a baseline proxy, not a separate measured DC channel. In the repeated participant-CV ablation, corrected-minus-raw MAE was -0.007 mmHg SBP (95% CI -0.042 to 0.029) and -0.030 mmHg DBP (-0.062 to 0.003). These intervals do not establish overall benefit or equivalence. The ablation tests the complete correction-and-feature-extraction configuration, not attenuation rescaling alone.

C. Calibration Dose and Anchoring

For participant p , let $b_p^{(k)}$ denote an anchor formed from exactly k initial cuff readings ($k \in \{1, 2, 3\}$), with $k > 1$ using their arithmetic mean. We define the session-anchored target

$$\Delta_{\text{session}}^{(k)}(p, i) = y_{p,i} - b_p^{(k)}, \quad (3)$$

where $y_{p,i}$ is the i -th subsequent cuff reading for participant p , and the prior-anchored target

$$\Delta_{\text{prior}}(p, i) = y_{p,i} - y_{p,i-1}^*, \quad (4)$$

where $y_{p,i-1}^*$ is the most recent valid prior measurement (which may itself be a tracked, non-calibration reading). These are not the same question. Eqn. 3 evaluates BP change after an initial calibration session without supplying later cuff values to the predictor. Eqn. 4 asks whether PPG can track change from the most recent cuff reading, which in this in-lab protocol can be only minutes old. A model evaluated under Eqn. 4 solves an easier problem; 83% of the historical fixed-split gain under prior anchoring was attributable to anchor replacement alone. We therefore lead with session-anchored tracking results and explicitly report calibration dose; neither anchoring scheme establishes cuff replacement. “One reading” refers only to $k = 1$; the primary $k = 3$ analysis is described as one-session calibration using three readings. The calibration-retention baseline predicts $\hat{\Delta} = 0$ under either anchor, i.e., “the BP has not changed since calibration (or since the last reading).” This is one baseline; the main context contrasts compare fitted models, and VitalDB additionally uses a calibration-plus-context control.

D. Context Operator

Let $c(p, i)$ collect acquisition context. We distinguish a wearable-state set (posture, activity, and optical signal quality) from a protocol-enriched set that additionally includes protocol phase, recording duration, and pressure-quality metadata. The wearable-state variables require corresponding sensing or classification at deployment; the protocol-enriched variables may encode AuroraBP acquisition structure and are therefore interpreted as an

internal upper bound on context utility. The context-augmented model is

$$\hat{\Delta}(p, i) = f_{\theta}(\Phi(x_{p,i}), \Phi(x_{p,i}) - \Phi(x_{p,\text{anchor}}), c(p, i)), \quad (5)$$

where f_{θ} is a gradient-boosted tree ensemble (HistGradientBoosting; hyperparameters grid-searched over learning rate, maximum leaf nodes, and ℓ_2 regularization) trained with either squared or absolute loss (Section IV), and the first-difference term $\Phi(x_{p,i}) - \Phi(x_{p,\text{anchor}})$ is the attractor-feature analogue of Eqs. 3–4 in feature space.

E. Calibration-Row Leakage Audit

A specific artifact threatens any context-augmented change-prediction model: a calibration-measurement indicator in $c(p, i)$ could identify trivial rows if the target reading were compared with itself; a calibration label alone does not guarantee this condition. A model that learns “calibration row $\Rightarrow$ predict no change” gains accuracy on those rows for free, in a way that would not transfer to genuine tracking of a new, non-calibration measurement. We formalize four checks:

- A Compare $|\Delta|$ on calibration-flagged vs. unflagged rows – are flagged rows trivially near-zero?
- B Re-score the same fitted models with flagged rows removed from the holdout only.
- C Refit with the calibration-flag feature removed, flagged rows still scored.
- D (Decisive) Refit with the flag feature removed and flagged rows excluded from both fitting and scoring – fit and evaluation restricted consistently.

Check D is decisive because A–C each leave one channel open (A is diagnostic only; B changes the scored population without changing the fitted model; C changes the model without changing the scored population). If the context gain measured under Check D remains close to the unaudited figure, the effect is not an artifact of the calibration flag. Supplementary Section B-F reports all four.

Finding 1. On this dataset, calibration-flagged rows are not near-zero-delta (mean $|\Delta| = 6.774$ vs. 9.902 mmHg SBP on development rows – smaller, but not trivially so), and Check D confirms the context gain survives essentially intact ($+0.614$ vs. the unaudited $+0.809$ mmHg SBP). Leave-one-out results implicate recording duration and post-exercise activity rather than the calibration flag (Supplementary Section B-F).

IV. Methodology

A. Datasets

AuroraBP (primary dataset) is the released Aurora-BP study resource [26], [30]. The files used here provide 1,221 participant records and 11,597 auscultatory measurements across in-lab calibration, static, exercise, and temporal-challenge protocol phases, with real Fitzpatrick skin-type labels (I: 311, II: 294, III: 138, IV: 44, V: 29, VI: 8 participants; 397 unlabeled). Waveforms were resolvable

TABLE I
AuroraBP Cohort and Strict-Endpoint Flow

Stage	Participants/measurements
Released participant records	1,221
Released auscultatory measurements	11,597
Participants with resolvable waveforms	672
Common three-reading calibration cohort	653
Matched post-initial-calibration rows	9,216
Later calibration-labelled rows excluded	1,337
Strict-endpoint scored rows	7,879

for 672 participants. The calibration-dose comparison uses the common subset of 653 participants with three valid initial calibration readings and 9,216 matched post-initial-calibration rows; the strict deployment endpoint excludes 1,337 later calibration-labelled rows, leaving 7,879 scored measurements. Historical ablations used a fixed 501/168 participant split (Fitzpatrick composition of the fixed holdout: I–II 133, III–IV 29, V–VI 6). Because that holdout was inspected repeatedly during model development, it is retained as historical development evidence rather than described as untouched validation. Four secondary sources appear in the development notebooks: BIDMC (53 subjects; 1,272 windows) [31]–[33], sampled MIMIC-II (1,500 windows) [34], [35], Welltory-PPG (106 windows) [36], and UCI (425 records; 3,268 windows) [37], [38]. BIDMC, MIMIC-II, and Welltory form a historical corrected-feature PCA bank; the primary candidate list does not include those PCA coordinates. UCI trains the BP-change prior that can enter the primary model. Synthetic pigmentation ranks on these auxiliary sources are simulation inputs, not measured participant traits, and support no equity claim.

Two additional datasets are used for independent follow-up analyses. The Appel–Theart Participant Study Data provide paired visible and infrared PPG, Monk Skin Tone (MST), and repeated pulse-rate references for 110 participant folders [39]. After waveform and metadata auditing, 108 participants contributed 630 analyzable pulse-rate windows, including 24 participants with MST 7–10. Each participant has only one cuff BP value; consequently BP analysis on this dataset is explicitly cross-sectional and does not test calibration tracking. VitalDB contains synchronized intraoperative waveforms and clinical tracks from 6,388 cases [40], [41]. The first 60 deterministic eligible-case positions yielded 55 development subjects. The next 240 positions yielded 228 eligible cases from 227 disjoint held-out subjects and 5,472 future windows after waveform/reference quality control. One held-out subject contributed two cases; each eligible case contributed 24 scored windows. Subject identities do not overlap development and holdout, and all cases belonging to a subject are grouped for inference. Only SNUADC/PLETH, Solar8000/ART_SBP, and Solar8000/ART_DBP were requested; ECG was neither loaded nor used. VitalDB has no skin-tone field. It evaluates future-window tracking within a separately developed intraoperative implementation, not pigmentation fairness or transfer of AuroraBP-fitted

parameters. Table II consolidates the analysis sample, role, and source references for every dataset used in the study.

B. Feature Extraction and Selection

AuroraBP features are extracted from the central 10-s optical window at $f_s = 500$ Hz (5,000 samples), using a third-order 0.5–8 Hz bandpass, $d = 4$, and $\tau = 5$ samples (10 ms). The extractor calculates $f_s = (N-1)/(t_{N-1} - t_0)$ from the recording timestamps; all 11,427 cached feature rows match the timestamp audit at 500 Hz within 2×10^{-11} Hz. The central waveform is not resampled before filtering. Sample entropy uses order 2, a base standardized tolerance of 0.2, and at most 600 samples; permutation entropy uses order 3. Recurrence analysis uses at most 300 standardized embedded points, Chebyshev distance, a base radius of 0.2 modified by the noise ratio, and diagonal lines of length at least 2. For the filtered AuroraBP signal, let $S(f)$ be its Welch power spectral density, with 4-s Hann segments and 50% overlap. The quality feature is

$$Q = \frac{\sum_{0.5 \leq f \leq 5} S(f)}{\sum_{0.1 \leq f \leq 15} S(f)}. \quad (6)$$

CSI is $0.4R_{\text{det}} + 0.3[1 - \text{clip}(H_{\text{RQA}}/5, 0, 1)] + 0.3Q$. Recalculation from the cached components agrees with stored CSI to 2.3×10^{-16} . This derived spectral ratio, stored as `raw_ppg_quality`, is distinct from AuroraBP’s supplied `optical_quality` context field and the independent pulse-fusion weights q_V, q_I . The historical hybrid calibration-anchored feature augmentation is denoted HCA below; its source-code identifiers retain the older name H1, distinct from hypothesis H1.

Within each outer training fold, the primary code ranks raw-feature differences and relative differences, elapsed-time variables, and the UCI prior after ridge residualization against baseline BP and log elapsed time. It applies `mutual_info_regression` to the residuals and retains up to 24 candidates plus baseline BP. SMD and AUC were historical diagnostics, not the primary selection algorithm. The name “no context” means no added context variables: baseline BP, candidate elapsed-time terms, and the UCI prior can still enter its shared core. No `ref_*` feature enters that core. The UCI prior is generated from corrected features, so absence of explicit Fitzpatrick terms does not prove absence of indirect Fitzpatrick dependence. The regenerated selection manifests retain the UCI prior in 26 of 30 endpoint/fold core models; all context arms share the corresponding selected core. Historical SD ratios are reported in Supplementary Table S4; they are not variance fractions.

C. Model and Training

For the primary AuroraBP comparisons, the supplied implementation uses median imputation followed by Hist-GradientBoosting with squared-error loss, learning rate 0.03, 15 maximum leaf nodes, ℓ_2 regularization 1.0, 300

iterations, and random seed 42. These settings are inherited from development rather than reselected using outer test folds. Row weights are inverse participant measurement counts, clipped at the training-set 2nd/98th percentiles and normalized to mean one. Participant folds are stratified by Fitzpatrick band, with seeds $42+26200+r$ for zero-based repeat r . The change gate is selected using fitted training predictions, not an independent inner validation set; reported activation is 1.0. This training-stage gate selection is an additional development limitation. Exploratory endpoint specialization uses a separate nested procedure.

D. Evaluation Protocol

Primary results use three repeats of five-fold participant-stratified outer cross-validation over the common 653-participant cohort. Every participant appears in an outer test fold once per repeat. Feature ranking, imputation, model fitting, and gate selection are repeated using only the corresponding outer training participants; no record or window from an outer test participant enters training. The final feature specification and context arms were frozen before this repeated analysis. Calibration-labelled rows are excluded from both fitting and scoring. Per-participant MAE is averaged across repeats, and 95% CIs are obtained by participant bootstrap (10,000 resamples); context contrasts use paired per-participant differences. This is repeated internal validation because AuroraBP informed earlier model development, not an external or prospective test. Historical fixed-split analyses use participant bootstrap (4,000 resamples) and are labelled accordingly. BHS and AAMI/ISO results are retained as indicative historical context, not compliance claims.

1) Claim hierarchy and multiplicity: The two primary retrospective comparisons are SBP and DBP wearable-state context versus the same core without added context. The rerun saves each participant’s mean MAE across the three repeats and forms paired no-context-minus-wearable-state contrasts. Two-sided paired sign-flip tests use the absolute mean contrast, 99,999 random sign assignments (seed 20260910), and $(b+1)/(B+1)$ Monte Carlo p values. Holm’s step-down procedure controls multiplicity across these two tests at $\alpha = 0.05$ [42]. Paired t tests are reported as a sensitivity analysis. No p value is reconstructed from a CI width. Primary contrasts additionally report 97.5% participant-bootstrap intervals, giving approximate Bonferroni simultaneous 95% coverage for the two endpoints; other intervals are marginal 95%. The sign-flip test assumes sign exchangeability under the paired null. Inference conditions on the fitted out-of-fold predictions, treats participants as the sampling units, and does not capture all dependence from shared training sets or earlier model development. Multiplicity correction does not make this retrospectively designated family prospective confirmation. All other comparisons are secondary or exploratory. The matched feature-family

TABLE II
Datasets Used in PC-AVCT and Their References

Dataset	Material analyzed	Role in this study	Source and access
AuroraBP	1,221 released participant records; 653 participants and 7,879 measurements in the strict primary endpoint	Primary session-calibrated BP evaluation, context arms, calibration dose, subgroup diagnostics, and feature-family ablation	Microsoft Research study and Zenodo dataset; available upon application under the provider's Data Use Agreement [26], [30]
BIDMC	53 subjects; 1,272 windows	Historical corrected-feature PCA representation; not a primary evaluation cohort	BIDMC PPG and Respiration Dataset, PhysioNet [31]–[33]
MIMIC-II	1,500 sampled windows	Historical corrected-feature PCA representation; not a primary evaluation cohort	MIMIC-II Clinical Database, PhysioNet [33]–[35]
Welltory-PPG	106 windows	Historical corrected-feature PCA representation; not a primary evaluation cohort	Published PPG signal collection [36]
UCI Cuff-Less BP	425 records; 3,268 windows	Training source for the BP-change prior that may enter AuroraBP models	UCI Machine Learning Repository [37], [38]
Appel–Theart	108 analyzed participants; 630 pulse-rate windows	Independent visible/infrared quality-fusion and guarded MST-correction analysis	Mendeley Data [39]
VitalDB	55 development subjects; 228 held-out cases from 227 subjects; 5,472 scored future windows	Independent intraoperative calibration-plus-context and context-plus-PPG BP tracking	VitalDB and PhysioNet [40], [41]

Counts describe the material used by this study and need not equal the full source repository. Auxiliary-source pigmentation ranks were synthetic simulation inputs; only AuroraBP supplied Fitzpatrick labels and Appel–Theart supplied Monk Skin Tone labels.

ablation is a secondary analysis on the same strict cohort and outer folds. Every arm uses baseline BP, elapsed-time terms, wearable-state context, the same fixed estimator, participant weights, and training-stage gate; none uses the UCI prior. The standard-PPG arm adds changes and relative changes in amplitude dispersion, skewness, kurtosis, spectral entropy, the band-power quality ratio, and pulse-arrival time. The attractor-augmented arm additionally adds sample entropy, permutation entropy, recurrence rate, determinism, RQA entropy, and CSI. Fixed semantic feature sets replace foldwise ranking for this ablation so the contrast tests the declared family addition. Paired participant-bootstrap intervals are marginal; sign-flip tests for the two attractor-versus-standard endpoints receive a separate Holm correction. Because several nonlinear complexity measures are not uniquely attractor-derived, this is evidence about the combined nonlinear/attractor augmentation rather than any single descriptor. For the fixed-estimator pigmentation diagnostic, the prior-anchored reference model was refitted on the original 501 development participants and evaluated on the same 168 held-out participants. Fitzpatrick bands stratify the resulting predictions; they do not alter the fitted estimator or its gates. The upstream corrected-feature UCI prior can introduce indirect Fitzpatrick dependence. MAE is averaged within participant before band means, 10,000-resample bootstrap intervals, and exploratory between-band contrasts are computed. A separate band summary uses the regenerated primary wearable-state predictions from all 653 participants. These diagnostics cannot establish fairness, particularly with only 6 and 25 V–VI participants in the respective cohorts. The later endpoint-specialization analyses are explicitly exploratory. They use the same 653-participant strict cohort and three repeats of five-fold participant outer validation, with three-fold

participant inner validation for candidate selection. For SBP, the candidates are ridge- or spline-based signed-residual corrections, shrunk and clipped using inner-fold choices; for DBP, the default candidate is the unchanged base predictor. The exploratory DBP router predicts whether $|\Delta_{\text{DBP}}|$ exceeds a training-fold-derived 80th-percentile cutoff. Its threshold is selected without outer-fold labels, and its output does not alter DBP: it only requests repeat measurement or recalibration. We report discrimination, calibration, warning precision and recall, and warning rate. Because these analyses followed inspection of earlier AuroraBP results, their confidence intervals quantify internal resampling stability rather than independent confirmation.

For the independent multi-wavelength mechanism analysis, visible and infrared pulse-rate estimates are weighted by within-window spectral concentration and autocorrelation quality. The tone-blind optical arm uses no MST information. The guarded arm permits an MST-dependent wavelength-weight exponent selected within four participant-disjoint inner folds. It activates only when the proposed nonzero exponent improves mean inner-fold MAE by at least 0.10 BPM and improves all inner folds; otherwise it returns exactly to tone-blind optical fusion. Evaluation uses 20 repeats of five-fold participant outer cross-validation. The supplied script averages repeated out-of-fold predictions for each window before calculating participant MAE; these are repeated-cross-fit ensemble estimates, not mean per-repeat errors. The audit's fixed equal-weight comparator is $(h_V + h_I)/2$, requires no fitting, and uses exactly the same 630 windows. Paired differences are averaged within each of 108 participants and bootstrapped 10,000 times (seed 20260910); those additional comparisons are post hoc with marginal intervals.

For VitalDB, the median BP and median features of

three early valid 20-s windows form the case-specific calibration anchor and are excluded from fitting and scoring; up to 24 later windows per case are scored. A calibration-plus-context control uses the anchor value, elapsed time, age, sex, and preoperative hypertension. The PPG arm adds changes in pulse rate, spectral concentration and entropy, waveform and derivative energy, distributional shape, beat-interval variability, and signal quality. After the initial 55-subject pilot, this stricter PPG-versus-calibration-plus-context comparison was frozen for the expanded test; it was not prospectively registered. Candidate estimators, regularization, clipping, and the context/PPG blend were selected using only the 55 development subjects. The selected models were then fitted on development data and applied once to 228 cases from 227 disjoint held-out subjects. Both endpoints selected a constant PPG weight of 0.5: the final prediction is the arithmetic mean of context and context-plus-PPG predictions. Signal quality is an input, not a window-dependent blend weight. This is a separately fitted implementation, not an unchanged AuroraBP model transferred to VitalDB. Reference quality control required plausible SBP/DBP, local arterial stability, and pulse pressure; it did not use prediction error. Contrasts were averaged within subject before a 10,000-resample participant bootstrap. The holdout was not used for scaling, model selection, clipping, or blend selection.

V. Experiments and Results

A. Primary Repeated Participant-Level Validation

Table III reports the refitted comparison on 653 participants and 7,879 strict-endpoint measurements. The wearable-state arm achieves 7.697/6.139 mmHg SBP/DBP. Its paired gains over no added context are 0.436 and 0.126 mmHg. The corresponding 97.5% bootstrap intervals are [0.305, 0.573] and [0.053, 0.200] mmHg. Holm-adjusted paired sign-flip p values are 0.00002 and 0.00013; paired t sensitivity tests also pass the two-endpoint Holm threshold. These conditional internal-validation results support the added context comparison. The optical-quality-only changes are negligible and directionally mixed: 0.022 mmHg lower SBP MAE but 0.008 mmHg higher DBP MAE than no added context, providing no meaningful evidence of incremental value from the supplied optical-quality field alone. Posture and activity are protocol annotations; deployment requires independent wearable estimates. The protocol-enriched arm additionally uses phase, duration, and pressure-quality metadata and is interpreted as an internal upper bound. As a direct consistency check, weighting the regenerated Primary-CV Fitzpatrick-band MAEs in Supplementary Table S8 by their participant counts (516/112/25) reproduces the Table III wearable-state estimates exactly before rounding: 7.6971776/6.1385779 mmHg.

B. Matched Feature-Family Ablation

Table IV reports the newly run secondary ablation. Adding the standard PPG descriptors to calibration,

TABLE III
Refitted Repeated Participant-Level Validation (653 Participants;
Three-Reading Anchor)

Arm	SBP MAE [95% CI]	DBP MAE [95% CI]
Calibration retention	9.790	6.767
No added context	8.133 [7.824, 8.455]	6.265 [6.019, 6.526]
Optical quality only	8.110 [7.802, 8.431]	6.273 [6.025, 6.535]
Wearable-state context	7.697 [7.407, 8.008]	6.139 [5.899, 6.391]
Protocol-enriched upper bound	7.523 [7.243, 7.815]	5.743 [5.527, 5.977]
Primary gain [97.5% CI]	0.436 [0.305, 0.573]	0.126 [0.053, 0.200]
Holm-adjusted paired p	0.00002	0.00013

TABLE IV
Secondary Matched AuroraBP Feature-Family Ablation (653
Participants)

Arm	SBP MAE [95% CI]	DBP MAE [95% CI]
Calibration, time, wearable context	7.920 [7.607, 8.256]	6.318 [6.067, 6.589]
+ Standard PPG descriptors	7.733 [7.441, 8.048]	6.137 [5.898, 6.388]
+ Nonlinear/attractor family	7.709 [7.418, 8.017]	6.128 [5.887, 6.384]
Attractor-family gain [95% CI]	0.025 [−0.018, 0.069]	0.009 [−0.033, 0.050]
Holm-adjusted paired p	0.532	0.662

time, and wearable context reduced participant MAE by 0.187 mmHg SBP (95% CI 0.088–0.286) and 0.180 mmHg DBP (0.096–0.266). Adding the nonlinear/attractor family to that standard arm produced only 0.025 mmHg SBP (−0.018 to 0.069) and 0.009 mmHg DBP (−0.033 to 0.050) further reductions. The Holm-adjusted paired sign-flip p values for those latter contrasts were 0.532 and 0.662. Thus the matched analysis supports an incremental contribution from the declared standard PPG descriptors but does not establish an attractor-family advantage.

C. Calibration Dose

Table V reports freshly refitted protocol-enriched models on identical participants and future measurements for each calibration dose. Ranking and fitting are repeated within the same participant folds. These dose gains belong to the protocol-enriched arm, not the lead wearable-state arm. Three readings are one-session repeated calibration, not one-reading calibration.

D. Prior-Anchored Isolation Ladder: Where the 26.2% Figure Really Comes From

Table VI decomposes the strongest raw number in this study – a 2.084 mmHg (26.2%) SBP reduction – into its components under a controlled, feature-set-fixed ladder. Eighty-three percent of that gain is attributable to re-anchoring alone (predicting from the most recent prior reading rather than the session calibration), which on an in-lab protocol can mean a reference only minutes old; this is a genuine tracking result, but it is not a cuffless-replacement result, and the two must not be reported interchangeably (Section III-C).

E. Secondary Pigmentation-Intervention Audit

Table VII compares four configurations, all built on the prior-anchored absolute-error+context base model of Table VI (chosen because it is the strongest available base

TABLE V
Refitted Protocol-Enriched Models by Calibration Dose (653 Participants)

Target	Readings	Retention	Model MAE [95% CI]	Gain vs. retention
SBP	1	10.547	8.257 [7.933, 8.602]	2.290 [2.051, 2.525]
SBP	2	10.037	7.750 [7.458, 8.055]	2.286 [2.066, 2.509]
SBP	3	9.790	7.523 [7.243, 7.815]	2.267 [2.044, 2.489]
DBP	1	7.213	6.280 [6.026, 6.548]	0.934 [0.739, 1.129]
DBP	2	6.900	5.873 [5.648, 6.112]	1.027 [0.833, 1.215]
DBP	3	6.767	5.743 [5.527, 5.977]	1.024 [0.832, 1.211]

TABLE VI
Historical Controlled Isolation Ladder, Prior-Anchored (SBP, common dev-only feature set, 168 holdout participants)

Step	MAE	Gain [95% CI]	Share
Session anchor (legacy)	7.956	—	—
→ Prior anchor	6.216	+1.739 [1.296, 2.197]	83%
→ Absolute-error loss	6.114	+0.102 [−0.028, 0.240]	5% [†]
→ +Context	5.795	+0.319 [0.189, 0.446]	15%
→ +HCA features	5.872	−0.077 [−0.183, 0.024]	−4% [†]
Total (session → full)		+2.084 [1.637, 2.533]	26.2%

[†] CI crosses zero: not established at 95%.

for isolating the marginal effect of skin-tone modeling): reference (no added Fitzpatrick terms; indirect dependence is possible), skin-features-only (continuous Fitzpatrick terms and interactions, no rebalancing), skin-features+balance (adds participant rebalancing across Fitzpatrick bands and group-specific shrinkage gates learned on development data), and updated-skin-aware (the full configuration). All four share participants, targets, holdout, base features, estimator, and hyperparameters. Skin features alone show no established benefit (5.770 vs. 5.795 mmHg SBP reference, not CI-distinguishable). Adding the combined rebalancing and group-gating intervention moves SBP MAE from 5.770 to 6.086 mmHg, a point difference of 0.316 mmHg. A paired interval for this specific incremental contrast is not supplied; the reported 0.291 mmHg interval instead compares the full intervention with the original reference arm. These experiments do not isolate rebalancing from group gating. In the signed contrast reported in Table VII, reference minus skin-aware is therefore −0.291 mmHg [−0.420, −0.170]. The combined intervention also widens the between-Fitzpatrick-band MAE gap, from 0.556 to 0.772 mmHg (SBP) and 0.721 to 0.940 mmHg (DBP). Per-band results suggest harm in the two better-represented bands; the Fitzpatrick V–VI comparison is indeterminate and rests on only six evaluation participants. These historical fixed-split results do not support adopting the tested combined intervention but do not establish general fairness properties.

The supplementary subgroup table now reports newly refitted historical-method estimates alongside the regenerated primary wearable-state results. The companion change record documents the differences from the superseeded subgroup values.

1) Repeated-CV pigmentation-correction intervention: Table VIII reports the historical V28 strict-cohort repeated-CV ablation retained from the development

TABLE VII
Historical Fixed-Split Skin-Tone Model Comparison (168 Evaluation Participants)

Configuration	SBP MAE	DBP MAE	vs. blind
Reference	5.795	4.486	—
Skin-features-only	5.770	4.477	Neutral
+ Demographic balance	6.086	4.712	Harm
Updated skin-aware (full)	6.086	4.712	Harm
Paired diff. (reference − full), mmHg [CI]; negative favors reference:			
SBP	−0.291 [−0.420, −0.170]		
DBP	−0.226 [−0.345, −0.116]		
Between-band gap, reference (SBP/DBP)	0.556 / 0.721 mmHg		
Between-band gap, skin-aware (SBP/DBP)	0.772 / 0.940 mmHg		

TABLE VIII
Pigmentation-Corrected vs. Raw Channel in Repeated Participant CV

Target	Scope	<i>n</i>	Raw	Corrected	Δ MAE [95% CI]
SBP	Overall	653	7.524	7.517	−0.007 [−0.042, 0.029]
SBP	I–II	516	7.487	7.482	−0.005 [−0.041, 0.032]
SBP	III–IV	112	7.683	7.620	−0.064 [−0.168, 0.036]
SBP	V–VI	25	7.571	7.793	0.222 [0.043, 0.401]
DBP	Overall	653	5.743	5.713	−0.030 [−0.062, 0.003]
DBP	I–II	516	5.681	5.639	−0.043 [−0.079, −0.005]
DBP	III–IV	112	6.064	6.077	0.014 [−0.066, 0.104]
DBP	V–VI	25	5.578	5.602	0.024 [−0.084, 0.131]

Δ = corrected minus raw; negative values favor correction.

record; it was not refitted in the V38 primary rerun. Neither overall interval establishes a benefit; equivalence was not tested. The V–VI SBP subgroup shows a CI-supported 0.222 mmHg worsening under correction, whereas the I–II DBP subgroup shows a small CI-supported improvement. These subgroup findings are heterogeneous and exploratory; they do not support a general skin-aware-model benefit. Adding explicit Fitzpatrick interactions to the corrected channel did not establish overall benefit: SBP Δ = 0.001 mmHg (95% CI −0.042 to 0.041) and DBP Δ = −0.037 mmHg (−0.076 to 0.002), relative to the raw model.

An exploratory signal diagnostic collapsed all repeated observations to one median per participant before inference. Fitzpatrick type was associated with lower raw-channel PPG quality ($\rho = -0.157$, 95% CI −0.234 to −0.079), greater normalized attractor dispersion ($\rho = 0.511$, 0.447 to 0.571), and lower recorded optical-quality score ($\rho = -0.174$, −0.253 to −0.092). Associations with optical noise ratio ($\rho = 0.032$, −0.044 to 0.112) and raw spectral entropy ($\rho = 0.032$, −0.046 to 0.106) did not exclude zero. These are Fitzpatrick-associated observed-signal differences, not measured-melanin effects; device gain, contact pressure, anatomy, and protocol may confound them.

Under the historical session-anchored C0-versus-C2 comparison, per-band SBP MAE decreases in Fitzpatrick I–II (7.865 → 7.169), III–IV (8.151 → 7.276), and nominally V–VI (9.042 → 6.280). The V–VI estimate is based on only six participants and is reported as indicative

TABLE IX
Independent Multi-Wavelength PPG Mechanism Test

Arm	Overall MAE (BPM)	MST 7–10 MAE
Visible only	4.366	5.434
Infrared only	5.037	5.165
Equal-weight fusion (post hoc)	3.985	4.587
Quality-weighted fusion	3.690	4.184
Guarded MST-aware fusion	3.703	4.265

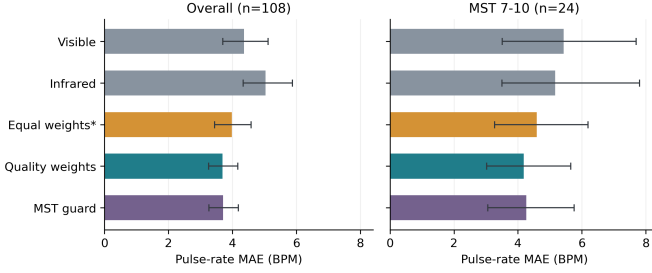


Fig. 1. Participant-balanced pulse-rate MAE with marginal 95% bootstrap intervals. The equal-weight comparison was added post hoc; quality weighting improves overall error relative to equal weighting, while guarded MST correction does not establish additional benefit.

only; it cannot support a fairness or parity claim.

F. Independent Quality-Fusion and Guarded-Correction Test

Table IX reports pulse-rate results on 108 participants. Quality-weighted fusion achieved 3.690 BPM MAE versus 4.366 for visible and 5.037 for infrared alone. A post-hoc audit of the same stored channel estimates added an equal-weight comparator, with 3.985 BPM MAE. Quality-minus-comparator paired differences were -0.676 BPM (95% CI -1.210 to -0.159) versus visible, -1.347 (-1.872 to -0.880) versus infrared, and -0.294 (-0.567 to -0.069) versus equal weighting. These marginal intervals support an overall pulse-rate benefit in this dataset; they do not establish BP utility. For MST 7–10, the quality-minus-equal-weight difference was -0.402 BPM (-0.929 to 0.010), which does not establish subgroup benefit. The exploratory MST guard activated in 5% of outer folds and achieved 3.703 BPM; its corrected-minus-tone-blind difference was 0.013 (-0.010 to 0.053), with no established overall benefit.

G. VitalDB Context-Plus-PPG Frozen Holdout

The expanded analysis used 55 development subjects to freeze the estimators and fixed blend, followed by one evaluation on 228 cases from 227 disjoint held-out subjects and 5,472 post-calibration windows. Table X separates the large gain over retaining the early calibration value from the stricter incremental PPG contrast. Calibration plus time/demographic context accounted for most of the improvement. The frozen fixed 50/50 blend reduced SBP MAE from 15.827 to 15.375 mmHg ($\Delta = -0.451$, 95% CI -0.781 to -0.121) and DBP MAE from 9.189 to 8.833 mmHg ($\Delta = -0.357$, -0.512 to -0.194). It

TABLE X
VitalDB Context-Plus-PPG Frozen Holdout (227 Subjects)

Target	Retention	Context	Context+PPG	PPG–Context [95% CI]
SBP	25.291	15.827	15.375	-0.451 [-0.781 , -0.121]
DBP	13.098	9.189	8.833	-0.357 [-0.512 , -0.194]

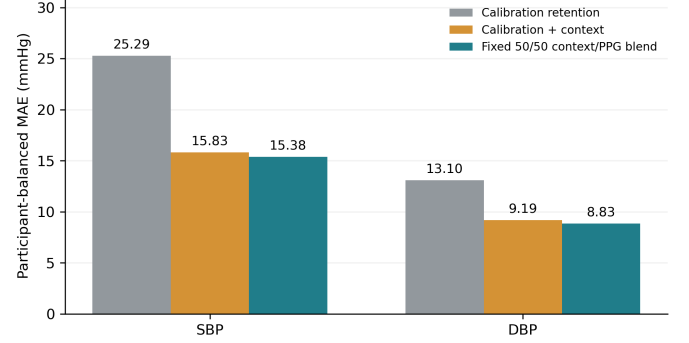


Fig. 2. VitalDB locked-holdout participant-balanced MAE. Calibration-plus-context explains most of the gain over retention; the frozen fixed context/PPG blend adds small CI-supported incremental improvements for both endpoints.

improved over context for 67.0% of subjects for SBP and 70.5% for DBP. For windows with ≥ 10 mmHg reference change, the incremental gains were 0.707 mmHg SBP (0.337–1.074) and 0.501 mmHg DBP (0.274–0.726). These results support a small incremental contribution of the PPG-augmented implementation within the VitalDB holdout, but absolute errors remain high and the test is retrospective and intraoperative rather than a prospective wearable validation.

H. State-of-the-Art Comparison

Table XI situates PC-AVCT against prior cuffless-BP results, organized around the axis we argue matters most for interpreting any number in this literature: what evaluation protocol produced it. Direct MAE comparison across datasets, populations, and BP ranges is only ever indicative – a point independently demonstrated by [1], which shows record-level leakage alone can inflate reported performance by more than 40 percentage points on the same data. We therefore report, for every comparator, whether the split was subject-independent, whether the setting was calibrated or calibration-free, and whether the study stratified by skin tone at all. Read narrowly, PC-AVCT’s wearable-state repeated-validation session-anchored MAE (7.697/6.139 mmHg) is higher than the best calibrated results reported elsewhere and DBP is higher than the calibration-free result of [8]. Direct ranking is not justified because datasets, populations, calibration doses, reference protocols, and evaluation procedures differ. AuroraBP includes substantial within-session physiological changes, and our analysis excludes calibration-labelled rows and separates participants throughout outer validation. The VitalDB test supplies an independent-dataset evaluation of a separately developed implementation through a dis-

TABLE XI
State-of-the-Art Comparison: Cuffless/Calibrated PPG Blood-Pressure Estimation

Method	Dataset (N)	Split	Calib.	SBP	DBP	Skin-tone report
PPG2BP-Net [43]	Intraop. (4,185 subj.)	Subject-indep.	Required	$0.209 \pm 7.509^*$	$0.150 \pm 4.549^*$	Not reported
Calibration-free CV-dynamics [8]	UCI/MIMIC-II (225 subj.)	Subject-indep.	Free	7.41	3.32	Not reported
Benchmark (representative baseline) [1]	Four public PPG-BP datasets	Subject-level CV	Mixed	15.60 (Sensors)	–	Not reported; quantifies leakage inflation up to >40 pp
AVCT (companion work) [5]	Smartphone PPG	Single-point calib.	1 reading	2.05	1.67	Not reported
rPPG equity field study [24]	Nigeria, Fitzpatrick V/VI (306 enrolled; 249 analyzable)	Independent cuff ref.	Free (rPPG)	15.4	10.9	Yes – Fitzpatrick V/VI
PC-AVCT wearable-state (this work), session-anchored	AuroraBP (653 subj.)	3×5-fold participant CV, bootstrap CI	3 readings	7.697 [7.407, 8.008]	6.139 [5.899, 6.391]	Real Fitzpatrick I–VI; V–VI underpowered
PC-AVCT protocol-enriched upper bound	AuroraBP (653 subj.)	Same internal CV	3 readings	7.523 [7.243, 7.815]	5.743 [5.527, 5.977]	Protocol metadata limit transportability
PC-AVCT context-plus-PPG holdout	VitalDB (227 held-out subj.)	Locked subject holdout	3 windows	15.375	8.833	Not available; temporal test only

* Reported as mean error $\pm$ SD, not MAE; not directly comparable to the MAE figures in other rows.

joint 227-subject holdout, but its intraoperative setting and high absolute errors still preclude clinical ranking or wearable-generalization claims. The appropriate conclusion is that the evidence is more stringently characterized, not that the method is superior to every comparator.

VI. Discussion

The primary AuroraBP comparison supports adding wearable-state context to the shared calibrated model, although posture and activity remain protocol annotations rather than independently estimated wearable states. The protocol-enriched arm achieves lower error but has greater dependence on acquisition metadata. Calibration dose materially affects that arm’s performance. Because the common core already includes baseline BP, elapsed-time candidates, and a UCI prior, these results do not isolate PPG utility beyond a calibration-plus-context-only model on AuroraBP.

The independent studies answer different questions. The post-hoc equal-weight comparison supports quality-weighted pulse-rate fusion on the multi-wavelength dataset. VitalDB supports a small incremental BP contribution from a separately fitted context-plus-PPG implementation, combined with context through a fixed 50/50 blend. It does not test quality-dependent blending or transfer of AuroraBP-fitted parameters. The combined evidence supports calibrated tracking research but does not establish one universally generalizing architecture, an attractor-specific advantage, or clinical interchangeability. Pigmentation interventions remain secondary; small samples, possible indirect feature dependence, preclude fairness conclusions.

Clinical accuracy boundary. The method remains a research prototype for calibrated trend tracking. On the historical measurement-level standards analysis, SBP

achieved BHS Grade C and failed the reported AAMI/ISO Criterion 1 check because error SD was 9.654 mmHg, above 8 mmHg. VitalDB absolute MAE was also high (15.375/8.833 mmHg). These failures cannot be repaired by emphasizing relative improvements, rejecting difficult observations after seeing their errors, or treating a calibrated tracking model as a validated cuff-replacement device. Existing data can support tail-error diagnostics and prospectively defined repeat-measurement rules, but not a claim of clinical interchangeability.

VII. Limitations

Small Fitzpatrick V–VI samples. The historical fixed holdout contains only 6 participants with Fitzpatrick V–VI labels; its fixed-model contrasts, Supplementary Table S8, and the historical per-band C0/C2 comparison are exploratory. The strict repeated-CV cohort includes 25 V–VI participants, which is larger but still small relative to I–II ($n = 516$) and insufficient for a definitive equity claim. Oversampling, duplicating, or synthetically weighting these participants cannot create independent evidence. Consequently, pigmentation modeling is a secondary intervention audit rather than a headline validation claim, and the equity question cannot be resolved using the current datasets. Synthetic pigmentation labels off AuroraBP. BIDMC, MIMIC, Welltory, and UCI contribute only a cross-dataset attractor-feature representation and prior; their pigmentation indices are permutation-derived synthetic ranks, not physiological measurements, and no equity claim in this paper rests on them. Single-site, in-lab protocol. AuroraBP measurements occur within < 1 h sessions in a controlled lab setting; the prior-anchored result (Section V-D) is therefore a same-session tracking result, not evidence about ambulatory, multi-day tracking, and we explicitly do not claim it as such. External evidence

remains retrospective. The repeated AuroraBP analysis estimates internal stability because AuroraBP informed earlier feature and model decisions. The Appel–Theart analysis is independent but validates pulse extraction rather than BP tracking. VitalDB supplies a frozen, different-site, different-device test on 227 held-out subjects, but it remains a retrospective intraoperative cohort with invasive BP, no skin-tone labels, and a different use setting from ambulatory wearables. It is not a prospective clinical validation. Adaptive pipeline development. Repeated cross-validation protects each fitted model from direct participant overlap, but it does not erase optimism from choosing feature families, context arms, and post-processing strategies after working extensively with AuroraBP. The endpoint-specific analyses are therefore labelled exploratory, and their bootstrap intervals are not substitutes for evaluation on a new cohort. Multiplicity and analytical breadth. Holm adjustment is applied to the two retrospectively designated primary paired tests. The remaining ablations and subgroup comparisons are exploratory, with marginal intervals. This correction addresses the stated two-endpoint family; it does not account for the full history of model-development choices or substitute for prospective registration. Calibration burden. The primary model averages three initial cuff readings. In the protocol-enriched dose experiment, exactly one reading produces higher MAE (8.257/6.280 mmHg); the corresponding wearable-state dose comparison is not reported. The three-reading result should be interpreted as one-session repeated calibration, not one-point calibration. Context transportability. The protocol-enriched model uses recording duration, phase, and pressure-quality metadata that may proxy AuroraBP protocol structure. The conservative wearable-state arm removes those variables and remains beneficial, but assumes that posture and activity are available from a deployable wearable classifier. Warning-router sensitivity. The DBP warning router identifies a small subset with high precision but recalls only 4.3% of extreme-change rows at its selected threshold. It cannot be used to claim comprehensive failure detection, clinical safety, or permission to omit confirmatory cuff measurement. Measurement-level vs. participant-level reporting. Because each participant contributes ~ 13 correlated measurements, uncertainty calculations that treat measurements as independent can be too narrow; measurement-weighted summaries also target a different estimand from participant-balanced summaries; we report both where possible and lead with participant-level bootstrap CIs for all primary claims. Pigmentation-correction operator is one design, not a proof of impossibility. Section III-B does not establish an overall benefit of the tested heuristic; its nonsignificant contrasts demonstrate neither equivalence nor impossibility of benefit. Quality fusion is not BP-fairness validation. The independent multi-wavelength result concerns pulse-rate extraction, and the available cuff value is cross-sectional. VitalDB has no pigmentation label. Neither result establishes that quality-aware fusion reduces BP error in Fitzpatrick V–VI

or MST 7–10 participants. Derived clinical categories, not a classifier. Any ACC/AHA-stage confusion analysis in the released figures is a post-hoc discretization of a regression model’s continuous output, not a trained classifier, and boundary-crossing small errors are penalized identically to large within-category errors. BHS/AAMI standards scope. Both grading systems were designed for cuff devices measuring absolute BP against an independent reference; applying them to a per-participant-calibrated change model is informative context, not compliance. Reproduction scope and feature attribution. All primary context and calibration-dose models were refitted from hashed feature-bank inputs in a recorded software environment, with outer-fold predictions, participant contrasts, feature selections, and gates saved. The historical fixed-estimator subgroup model was also refitted, and its new estimates replace the inherited subgroup values. The analysis rerun begins at feature-bank inputs; it does not repeat raw-waveform extraction. The UCI prior uses corrected features, so “no added Fitzpatrick terms” does not imply complete pigmentation independence. The newly executed matched ablation isolates the declared nonlinear/attractor feature-family addition from standard PPG descriptors under a shared AuroraBP protocol; it does not establish a benefit for that addition or isolate any one descriptor.

VIII. Conclusion

PC-AVCT provides retrospective evidence for context-augmented session-calibrated BP tracking. The AuroraBP primary comparison supports added wearable-state context; the calibration-dose result belongs to the protocol-enriched arm. A matched secondary ablation supports an incremental contribution from standard PPG descriptors but does not establish additional benefit from the combined nonlinear/attractor family. A separate post-hoc pulse-rate comparison supports quality-weighted fusion over equal weighting. Within VitalDB, a separately fitted context-plus-PPG implementation and fixed 50/50 blend provide small incremental BP improvements beyond calibration, time, and demographics, with high absolute errors. These results do not isolate quality-dependent BP blending, prove attractor-specific benefit, demonstrate transfer of the fitted AuroraBP model, validate AVCT/CST/ADT as physiological laws, or establish equitable, prospective, ambulatory, standards-compliant, or cuff-replacement performance.

Code and Reproduction Availability

The companion reproducibility package is publicly available on GitHub:

<https://github.com/farouze/pc-avct-reproducibility>

It contains executable rerun and statistical-analysis scripts, the matched feature-family ablation, preserved V26 helper functions, pinned dependencies, input SHA-256 hashes, model-selection manifests, regenerated aggregate outputs, and sanitized copies of the author-supplied

development notebooks. Participant-level data, row predictions, and paired participant contrasts are excluded from the public repository under the dataset access conditions and retained locally for audit. The versioned GitHub release is the software archive cited here; the dataset DOI below identifies data, not this implementation.

Data Availability

AuroraBP is available upon application through the Zenodo record cited in [30] (DOI: 10.5281/zenodo.19099166) under the provider's Data Use Agreement. BIDMC and MIMIC-II are available through PhysioNet, and the UCI Cuff-Less Blood Pressure Estimation dataset is available through the UCI Machine Learning Repository. The multi-wavelength participant data are available through Mendeley Data [39]; VitalDB is openly available through PhysioNet [41]. Access conditions and citations are listed in Section IV-A. No directly identifying participant data are included in this manuscript or its figures.

Ethics Statement

This study is a secondary computational analysis of previously collected, deidentified datasets. The authors recruited no participants, performed no intervention, had no participant contact, received no direct identifiers, and made no attempt to reidentify participants. Morgan State University's published IRB process states that submission is not required for research using deidentified data or publicly available datasets [44]. Accordingly, no additional institutional ethics review or new participant consent was required for this secondary analysis. Ethics approvals and consent procedures for the original data collections are reported by the respective dataset providers and source publications [26], [30], [32], [39]–[41].

Author Contributions

Farouk Ganiyu Adewumi: conceptualization, methodology, software, validation, formal analysis, data curation, visualization, and writing of the original draft. Timothy Oladunni: conceptualization, methodology, supervision, and review and editing of the manuscript. Both authors approved this revised manuscript and the public code release.

Competing Interests

The authors declare no competing financial or non-financial interests.

Funding

This work was supported in part by Morgan State University's Office of Research and Graduate Programs.

Acknowledgment

We thank the participants, investigators, and data stewards who made the secondary datasets used in this study available.

References

- [1] S. González, W.-T. Hsieh, and T. P.-C. Chen, "A benchmark for machine-learning based non-invasive blood pressure estimation using photoplethysmogram," *Scientific Data*, vol. 10, p. 149, 2023.
- [2] J. Kostrna, E. Oparina, C. Palacios, A. J. Rodriguez, J. Pei, A. Ajmal, and J. C. Ramella-Roman, "PPG-based heart rate accuracy in diverse populations: Investigating inequities across body composition and skin tones," *arXiv preprint arXiv:2601.22377*, 2026.
- [3] S. Singh, M. R. Bennett, C. Chen, S. Shin, H. Ghanbari, and B. W. Nelson, "Impact of skin pigmentation on pulse oximetry blood oxygenation and wearable pulse rate accuracy: Systematic review and meta-analysis," *Journal of Medical Internet Research*, vol. 26, p. e62769, 2024.
- [4] L. Osorio-Sanchez, J. M. May, and P. A. Kyriacou, "Evaluation of skin pigmentation effect on photoplethysmography signals using a vascular finger phantom with tunable optical and mechanical properties," *Journal of Biomedical Optics*, vol. 30, no. 11, p. 117002, 2025.
- [5] T. Oladunni and F. G. Adewumi, "Attractor-vascular coupling theory: Formal grounding and empirical validation for AAMI-standard cuffless blood pressure estimation from smartphone photoplethysmography," *arXiv preprint arXiv:2605.10871*, 2026.
- [6] M. Elgendi, R. Fletcher, Y. Liang, N. Howard, N. H. Lovell, D. Abbott, K. Lim, and R. Ward, "The use of photoplethysmography for assessing hypertension," *npj Digital Medicine*, vol. 2, p. 60, 2019.
- [7] R. Mukkamala, J.-O. Hahn, O. T. Inan, L. K. Mestha, C.-S. Kim, H. Töreyn, and S. Kyal, "Toward ubiquitous blood pressure monitoring via pulse transit time: Theory and practice," *IEEE Transactions on Biomedical Engineering*, vol. 62, no. 8, pp. 1879–1901, 2015.
- [8] H. Samimi and H. R. Dajani, "A PPG-based calibration-free cuffless blood pressure estimation method using cardiovascular dynamics," *Sensors*, vol. 23, no. 8, p. 4145, 2023.
- [9] C.-F. Wang, T.-Y. Wang, P.-H. Kuo, H.-L. Wang, S.-Z. Li, C.-M. Lin, S.-C. Chan, T.-Y. Liu, Y.-C. Lo, S.-H. Lin, and Y.-Y. Chen, "Upper-arm photoplethysmographic sensor with one-time calibration for long-term blood pressure monitoring," *Biosensors*, vol. 13, no. 3, p. 321, 2023.
- [10] M. Elgendi, F. Haugg, R. R. Fletcher, J. Allen, H. Shin, A. Alian, and C. Menon, "Recommendations for evaluating photoplethysmography-based algorithms for blood pressure assessment," *Communications Medicine*, vol. 4, p. 140, 2024.
- [11] F. Takens, "Detecting strange attractors in turbulence," in *Dynamical Systems and Turbulence*, Warwick 1980, ser. Lecture Notes in Mathematics. Springer, 1981, vol. 898, pp. 366–381.
- [12] J.-P. Eckmann, S. O. Kamphorst, and D. Ruelle, "Recurrence plots of dynamical systems," *Europhysics Letters*, vol. 4, no. 9, pp. 973–977, 1987.
- [13] N. Marwan, M. C. Romano, M. Thiel, and J. Kurths, "Recurrence plots for the analysis of complex systems," *Physics Reports*, vol. 438, no. 5–6, pp. 237–329, 2007.
- [14] J. S. Richman and J. R. Moorman, "Physiological time-series analysis using approximate entropy and sample entropy," *American Journal of Physiology-Heart and Circulatory Physiology*, vol. 278, no. 6, pp. H2039–H2049, 2000.
- [15] C. Bandt and B. Pompe, "Permutation entropy: A natural complexity measure for time series," *Physical Review Letters*, vol. 88, no. 17, p. 174102, 2002.
- [16] J. Allen, "Photoplethysmography and its application in clinical physiological measurement," *Physiological Measurement*, vol. 28, no. 3, pp. R1–R39, 2007.
- [17] C. Hörandtner, M. Bachler, W. Sehnert, I. Mikisek, T. Mengden, S. Wassertheurer, and C. C. Mayer, "Attractor reconstruction for quantifying the arterial pulse wave morphology during device-guided slow breathing," *Cardiovascular Engineering and Technology*, vol. 13, no. 6, pp. 939–949, 2022.
- [18] T. Oladunni and F. G. Adewumi, "Cardiac stability theory: An axiomatically grounded framework for continuous cardiac health monitoring via smartphone photoplethysmography," *arXiv preprint arXiv:2604.23876*, 2026.
- [19] T. Oladunni and F. G. Adewumi, "Attractor domain theory: A mathematical framework for cardiovascular attractor analysis

- with wearable photoplethysmography (PPG) validation,” arXiv preprint arXiv:2606.22039, 2026.
- [20] C. E. Shannon, “A mathematical theory of communication,” *Bell System Technical Journal*, vol. 27, no. 3, pp. 379–423, 1948.
 - [21] T. M. Cover and J. A. Thomas, *Elements of Information Theory*, 2nd ed. Wiley-Interscience, 2006.
 - [22] R. Al-Halawani, M. Qassem, and P. A. Kyriacou, “Analysis of the effect of skin pigmentation and oxygen saturation on monte carlo-simulated reflectance photoplethysmography signals,” *Sensors*, vol. 25, no. 2, p. 372, 2025.
 - [23] R. Jakachira, W. Yan, S. C. Thomas, Y. Macias-Sanchez, L. Werner, J. A. Burrow, S. Dunsiger, and K. C. Toussaint, Jr., “Evaluation of a polarization-sensitive, dual-wavelength wearable photoplethysmography sensor across a range of skin tones,” *Biophotonics Discovery*, vol. 3, no. 1, p. 012509, 2026.
 - [24] D. Dasa and P. Davies, “Skin tone and diagnostic equity in contactless blood pressure screening: A prospective observational field evaluation of remote photoplethysmography in nigeria,” *BMJ Open*, vol. 16, no. 6, p. e119311, 2026.
 - [25] C. MacIsaac, M. Nguyen, A. Uy, T. Kong, and A. Hedayatipour, “A programmable gain calibration method to mitigate skin tone bias in PPG sensors,” *Biosensors*, vol. 15, no. 7, p. 423, 2025.
 - [26] R. Mieloszyk, H. Twede, J. Lester, J. Wander, S. Basu, G. Cohn, G. Smith, D. Morris, S. Gupta, D. Tan, N. Villar, M. Wolf, S. Malladi, M. Mickelson, L. Ryan, L. Kim, J. Kepple, S. Kirchner, E. Wampler, R. Terada, J. Robinson, R. Paulsen, and T. S. Saponas, “A comparison of wearable tonometry, photoplethysmography, and electrocardiography for cuffless measurement of blood pressure in an ambulatory setting,” *IEEE Journal of Biomedical and Health Informatics*, vol. 26, no. 7, pp. 2864–2875, 2022.
 - [27] I. Sanches, V. V. Gomes, C. Caetano, L. S. B. Cabrera, V. H. Cene, T. Beltrame, W. Lee, S. Baek, and O. A. B. Penatti, “MIMIC-BP: A curated dataset for blood pressure estimation,” *Scientific Data*, vol. 11, p. 1233, 2024.
 - [28] F. M. Dias, D. A. C. Cardenas, M. A. F. Toledo, F. A. C. Oliveira, E. Ribeiro, J. E. Krieger, and M. A. Gutierrez, “Exploring the limitations of blood pressure estimation using the photoplethysmography signal,” *Physiological Measurement*, vol. 46, no. 4, p. 045007, 2025.
 - [29] S. Mehta, N. Kwatra, M. Jain, and D. McDuff, “Examining the challenges of blood pressure estimation via photoplethysmogram,” *Scientific Reports*, vol. 14, p. 18318, 2024.
 - [30] Microsoft Research, “Aurora-BP Study Dataset,” Zenodo, 2022, available upon application under the provider’s Data Use Agreement.
 - [31] M. A. F. Pimentel, A. E. W. Johnson, P. H. Charlton, and D. A. Clifton, “BIDMC PPG and Respiration Dataset,” *PhysioNet*, 2018, version 1.0.0.
 - [32] M. A. F. Pimentel, A. E. W. Johnson, P. H. Charlton, D. Birrenkott, P. J. Watkinson, L. Tarassenko, and D. A. Clifton, “Toward a robust estimation of respiratory rate from pulse oximeters,” *IEEE Transactions on Biomedical Engineering*, vol. 64, no. 8, pp. 1914–1923, 2017.
 - [33] A. L. Goldberger, L. A. N. Amaral, L. Glass, J. M. Hausdorff, P. C. Ivanov, R. G. Mark, J. E. Mietus, G. B. Moody, C.-K. Peng, and H. E. Stanley, “PhysioBank, PhysioToolkit, and PhysioNet: Components of a new research resource for complex physiologic signals,” *Circulation*, vol. 101, no. 23, pp. e215–e220, 2000.
 - [34] M. Saeed, M. Villarroel, A. T. Reisner, G. Clifford, L.-W. Lehman, G. Moody, T. Heldt, T. H. Kyaw, B. Moody, and R. G. Mark, “Multiparameter intelligent monitoring in intensive care ii (MIMIC-II): A public-access intensive care unit database,” *Critical Care Medicine*, vol. 39, no. 5, pp. 952–960, 2011.
 - [35] M. Saeed, M. Villarroel, A. T. Reisner, G. Clifford, L.-W. Lehman, G. Moody, T. Heldt, T. Kyaw, B. Moody, and R. Mark, “MIMIC-II Clinical Database,” *PhysioNet*, 2011, version 2.6.0.
 - [36] A. Neshitov, K. Tyapochkin, E. Smorodnikova, and P. Pravdin, “Wavelet analysis and self-similarity of photoplethysmography signals for HRV estimation and quality assessment,” *Sensors*, vol. 21, no. 20, p. 6798, 2021.
 - [37] M. Kachuee, M. M. Kiani, H. Mohammadzade, and M. Shabany, “Cuff-Less Blood Pressure Estimation,” *UCI Machine Learning Repository*, 2015.
 - [38] M. Kachuee, M. M. Kiani, H. Mohammadzade, and M. Shabany, “Cuff-less high-accuracy calibration-free blood pressure estimation using pulse transit time,” in *2015 IEEE International Symposium on Circuits and Systems (ISCAS)*, 2015, pp. 1006–1009.
 - [39] A. Appel and R. Theart, “Photoplethysmography device data: Hardware and participant study insights,” *Mendeley Data*, 2025.
 - [40] H.-C. Lee, Y. Park, S. B. Yoon, S. M. Yang, D. Park, and C.-W. Jung, “VitalDB, a high-fidelity multi-parameter vital signs database in surgical patients,” *Scientific Data*, vol. 9, p. 279, 2022.
 - [41] H.-C. Lee and C.-W. Jung, “VitalDB, a high-fidelity multi-parameter vital signs database in surgical patients,” *PhysioNet*, 2022, version 1.0.0.
 - [42] S. Holm, “A simple sequentially rejective multiple test procedure,” *Scandinavian Journal of Statistics*, vol. 6, no. 2, pp. 65–70, 1979.
 - [43] J. Joung, C.-W. Jung, H.-C. Lee, M.-J. Chae, H.-S. Kim, J. Park, W.-Y. Shin, C. Kim, M. Lee, and C. Choi, “Continuous cuffless blood pressure monitoring using photoplethysmography-based PPG2BP-net for high intrasubject blood pressure variations,” *Scientific Reports*, vol. 13, p. 8605, 2023.
 - [44] Morgan State University, “IRB Process Synopsis,” Office of Research Administration, 2023, accessed: 2026-09-10. [Online]. Available: https://www.morgan.edu/Documents/ADMINISTRATION/OFFICES/ora/IRB%20Forms/2023-03-21_IRB_Process_Synopsis.pdf